

ULLADULLA, Australia

WMO: 949380

Lat: 35.3636S

Lon: 150.4828E

Elev: 120

StdP: 14.63

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.5	46.0	26.6	20.7	55.5	29.2	23.3	55.2	27.8	55.2	23.5	55.5	6.8	260	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.0	83.0	65.1	78.6	66.0	76.3	67.3	72.2	76.0	71.1	74.8	70.1	73.9	10.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	114.9	74.0	69.9	110.4	73.4	68.8	106.1	72.6	36.1	76.1	35.1	75.0	34.2	74.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	21.3	18.4	DB	41.3	95.1	1.7	5.6	40.1	99.2	39.1	102.5	38.1	105.7	36.9	109.7	(4)
(5)			WB	36.2	74.2	1.5	1.8	35.1	75.4	34.2	76.5	33.4	77.5	32.3	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	68.8	69.0	67.4	63.5	59.5	55.9	54.5	55.6	58.4	61.2	63.4	66.4	(6)	
	DBStd	6.38	3.74	3.47	3.53	3.79	3.66	3.11	3.25	3.76	4.68	5.12	4.59	4.07	(7)	
	HDD50	5	0	0	0	0	0	1	3	1	0	0	0	0	(8)	
	HDD65	1625	7	5	16	70	174	274	326	292	206	142	83	30	(9)	
	CDD50	4360	582	533	540	405	296	177	141	175	253	348	402	508	(10)	
	CDD65	506	124	118	91	25	5	0	0	1	9	26	34	73	(11)	
	CDH74	1292	281	235	155	59	8	0	1	4	56	183	130	180	(12)	
	CDH80	303	66	40	26	12	0	0	0	0	9	64	42	44	(13)	
(14) Wind	WSAvg	8.3	8.7	8.6	8.4	8.0	7.6	8.1	7.8	8.3	8.5	8.5	8.6	8.5	(14)	
(15) Precipitation	PrecAvg	49.4	3.7	5.5	5.2	4.7	4.3	4.8	2.4	2.8	3.2	4.2	4.5	3.9	(15)	
	PrecMax	84.6	9.8	25.6	17.0	19.5	14.2	26.5	7.7	15.5	10.5	15.6	19.7	12.9	(16)	
	PrecMin	22.5	0.3	0.4	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)	
	PrecStd	16.1	2.4	5.2	4.3	4.5	3.3	5.0	2.0	3.4	2.4	3.4	3.5	3.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	85.1	82.9	80.3	75.0	68.2	69.0	73.3	80.5	88.5	86.2	86.0	(19)	
		MCWB	66.6	69.3	67.1	61.4	58.8	56.7	53.2	54.6	58.4	62.2	62.5	65.4	(20)	
	2%	DB	78.8	78.3	77.2	74.6	70.4	65.4	64.8	67.8	73.8	79.0	76.6	78.5	(21)	
		MCWB	69.0	70.6	68.1	63.2	59.6	56.0	52.9	54.0	56.8	59.5	63.4	66.8	(22)	
	5%	DB	76.3	76.4	75.1	72.1	68.1	63.4	62.6	64.7	69.2	72.7	72.6	74.7	(23)	
		MCWB	69.4	69.9	68.0	63.7	59.4	55.1	52.4	53.8	56.3	59.5	64.6	67.1	(24)	
	10%	DB	74.6	74.8	73.5	70.3	66.2	61.8	60.6	62.4	66.0	68.6	70.1	72.7	(25)	
		MCWB	68.5	69.0	67.2	63.1	58.4	54.1	51.8	53.3	56.7	60.1	63.9	66.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	73.8	73.0	69.0	64.9	60.9	58.8	60.0	63.3	66.3	69.1	72.2	(27)	
		MCDB	77.1	78.2	75.9	72.9	67.7	65.3	62.4	64.7	69.8	74.2	74.0	77.2	(28)	
	2%	WB	71.9	72.4	71.2	67.3	63.1	58.5	56.6	58.1	61.5	64.5	67.4	70.2	(29)	
		MCDB	75.6	76.3	74.5	71.3	66.9	62.4	61.0	62.8	66.7	70.0	72.1	74.2	(30)	
	5%	WB	70.9	71.2	69.9	65.9	61.5	57.1	54.9	56.6	60.0	63.0	66.0	68.9	(31)	
		MCDB	74.7	74.8	73.3	70.1	66.1	61.1	59.4	61.7	65.2	68.4	70.6	72.7	(32)	
	10%	WB	69.7	70.1	68.5	64.4	59.8	55.7	53.4	55.0	58.4	61.8	64.7	67.5	(33)	
		MCDB	73.6	73.8	72.3	68.8	64.9	60.1	58.3	60.7	64.2	67.2	68.9	71.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	10.0	10.6	11.6	11.5	10.7	11.7	12.7	13.2	13.1	11.5	10.8	(35)	
		MCDBR	13.0	12.4	13.0	14.3	13.7	12.7	13.9	15.4	19.3	21.9	17.1	15.1	(36)	
	5% WB	MCWBR	7.0	7.1	8.0	9.1	8.7	8.1	8.4	8.8	9.2	9.3	8.4	7.9	(37)	
		MCWBR	10.8	11.2	10.6	12.2	11.5	10.3	11.4	12.9	14.2	14.5	13.0	11.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.408	0.381	0.354	0.327	0.311	0.299	0.292	0.302	0.333	0.351	0.380	0.382	(40)	
	taud		2.400	2.512	2.569	2.619	2.611	2.641	2.654	2.618	2.525	2.505	2.455	2.469	(41)	
	Ebn at Noon		294	296	292	283	271	266	276	288	294	302	301	304	(42)	
	Edh at Noon		40	34	30	25	23	20	21	25	31	34	37	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2005	1723	1482	1177	934	723	862	1132	1525	1793	1889	2049	(44)	
	RadStd		166	142	107	100	52	61	70	64	95	132	152	143	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-2	-148	+310	+144

Nomenclature: See separate page